

ALLIANCE UNIVERSITY

NAAC GRADE A+ ACCREDITED UNIVERSITY

School of Advanced Computing

Department of Computer Science

MCA. General,

E1CAB703 Computer Networks and Communications

Mini Project Report

Project Title:

Network Security Management System Using Cisco Packet Tracer

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Date of Submission:

27-08-2026

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CERTIFICATE

This is to certify that the Mini Project work entitled **"Network Security Management System Using Cisco Packet Tracer"** is the bonafide work done by **Prajwal C K (2511022250087)**, submitted for the internal mark requirements for the course **E1CAB703 – Computer Networks and Communications**, Master of Computer Applications (MCA), School of Advanced Computing, Department of Computer Science, during the academic year **2025–2026**.

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1. INTRODUCTION

Network security management has become one of the most critical challenges faced by modern organizations, educational institutions, enterprises, and government bodies that depend on interconnected computer systems for daily operations. As organizations grow, their networks typically expand to include multiple departments, remote users, wireless devices, and externally facing servers, each of which introduces additional attack surface. Traditional flat network designs, in which all devices share a single broadcast domain with little to no internal segmentation, are particularly vulnerable because a single compromised host can potentially communicate with, scan, or attack every other device on the network.

This project, titled "Network Security Management System Using Cisco Packet Tracer," presents the design, configuration, and simulation of a secure enterprise network. Cisco Packet Tracer is used to build a segmented network consisting of a Layer 3 core switch, an ISP-facing router, a dedicated firewall appliance, multiple department-level VLANs (Management, HR, Finance, and IT), a dedicated server VLAN hosting DHCP and web services, a wireless access router, and a set of external devices representing untrusted outside hosts.

Rather than relying on a single perimeter defense, the proposed design applies defense-in-depth: VLAN-based segmentation isolates department traffic, a firewall filters traffic entering the internal network from the ISP router, static and access-controlled IP addressing restricts which devices can reach sensitive services, and centralized DHCP and web/HTTP-HTTPS services are hosted on a hardened server segment. The project demonstrates how these layers work together to reduce the attack surface, contain the impact of a compromised host, and provide administrators with a manageable and auditable network structure.

1.1 Problem Statement

Organizations that operate flat, unsegmented networks face a high risk of lateral movement by attackers, unauthorized inter-department access, rogue DHCP servers, unsecured management interfaces, and uncontrolled traffic between trusted internal hosts and untrusted external hosts. There is a need for a structured network security management design that segments traffic by department and function, filters traffic at trust boundaries using a firewall, secures core services such as DHCP and web hosting, and enforces consistent IP addressing and access policies across the network.

1.2 Objectives

- To design and simulate a segmented enterprise network using Cisco Packet Tracer.
- To isolate department traffic (Management, HR, Finance, and IT) using VLANs on a Layer 3 core switch.
- To deploy a dedicated firewall appliance between the ISP router and the internal network to filter untrusted traffic.
- To configure a centralized server segment (VLAN 99) providing DHCP addressing and HTTP/HTTPS web services.
- To apply static IP addressing and interface-level security configuration on network devices.
- To secure wireless connectivity for mobile/laptop clients using an access router.

- To verify inter-VLAN connectivity, service reachability, and boundary filtering through simulation testing.
- To demonstrate how layered network security controls reduce the overall attack surface of an enterprise network.

1.3 Scope of the Project

The scope of this project is to design and simulate a multi-VLAN enterprise network consisting of a core Layer 3 switch, an ISP router, a firewall, four department access switches (Management, HR, Finance, IT), a dedicated server VLAN, a wireless router, and a small set of external devices representing outside connections. The project focuses on network segmentation, boundary filtering, centralized service hosting, and IP address management. The implementation is carried out entirely as a simulation in Cisco Packet Tracer and provides a foundation for extending the design with more advanced security controls such as intrusion detection/prevention systems, Zero Trust access policies, SIEM-based monitoring, and automated firewall rule auditing.

2.Literature Survey

Topic: Network Security Management

This paper presents a simulation-based study that segments a flat internal network into IT, Finance, and Management VLANs combined with Access Control Lists (ACLs) to restrict inter-department traffic. Using Cisco Packet Tracer and router-on-a-stick inter-VLAN routing, an Extended ACL is applied on the Finance VLAN sub-interface so that the Finance department cannot reach the IT or Management VLANs, while the IT department retains full access to Finance for support purposes. [1].

Dataset: Simulated internal network topology (IT and Finance case study) built in Cisco Packet Tracer

Modules: VLAN Segmentation, Router-on-a-Stick Inter-VLAN Routing, Extended ACL Policy Enforcement, Connectivity Testing

Algorithm: Extended Access Control List filtering combined with VLAN-based broadcast-domain segmentation

Accuracy: Not explicitly quantified; validated through successful/blocked ping tests between VLANs

Research Gap: The design is evaluated only on a small three-VLAN topology and does not examine ACL rule scaling or misconfiguration risk in larger multi-department networks.

This case study explores the design and simulation of Virtual Local Area Networks using Cisco

Packet Tracer, focusing on how VLANs segment a complex network into smaller broadcast domains for improved manageability and security. The work highlights Packet Tracer's simulation and visualization capabilities for teaching switching concepts and examines security aspects of VLAN design, including issues that can arise with VLAN Trunking Protocol configuration. [2].

Dataset: Case-study network topology simulated entirely within Cisco Packet Tracer (no external dataset)

Modules: VLAN Creation, Trunking Configuration, VTP Domain Setup, Routing Between VLANs

Algorithm: IEEE 802.1Q VLAN tagging with VTP-based configuration propagation

Accuracy: Not applicable; the study is descriptive/demonstrative rather than quantitatively evaluated

Research Gap: The paper notes that VTP misconfiguration issues are under-researched but does not itself propose or test a mitigation mechanism for VTP-related failures.

This paper proposes a secure campus network design that combines multiple VLANs with critical security configurations to protect data integrity and reliability during transmission between end users. Implemented and validated using Cisco Packet Tracer, the design demonstrates how a layered combination of switching, routing, and access-control measures can build a stable and secure campus community network. [3].

Dataset: Campus network topology with multiple VLANs modeled in Cisco Packet Tracer

Modules: VLAN Configuration, Secure Switch Port Setup, Inter-VLAN Routing, Data-Flow Monitoring

Algorithm: Rule-based network segmentation with layered security configuration on switches and routers

Accuracy: Not explicitly quantified; assessed through functional connectivity and security-configuration verification

Research Gap: The study does not benchmark the proposed configuration against a baseline unsegmented network to quantify the actual reduction in attack surface.

This paper analyzes the working mechanism of firewall technology and designs a preliminary firewall system for a communications department based on the IPX2400 processor, implementing its packet-filtering function to defend against foreign intruders who could otherwise extract sensitive information using conventional firewalls alone. [4].

Dataset: Hardware-oriented firewall prototype design (processor-level architecture, no public dataset)

Modules: Packet Filtering Engine, Hardware Firewall Architecture, IPX2400-based Processing Unit

Algorithm: Rule-based packet filtering implemented at the processor/hardware level

Accuracy: Not explicitly quantified; validated through description of the packet-filtering implementation

Research Gap: The design is discussed at an architectural level without performance benchmarking (throughput, latency) or comparison against modern software-defined firewalls.

This paper presents an eight-building campus area network in which end devices are divided into multiple VLANs for security, and additional security approaches are deployed on the switches and routers that interconnect the buildings. The study demonstrates blocking of unauthorized ping requests using a combination of VLANs and ACLs and discusses the role of key Layer-2 protocols in protecting the overall campus network. [5].

Dataset: Eight-building campus network topology simulated in Cisco Packet Tracer

Modules: Multi-Building VLAN Segmentation, ACL-based Traffic Filtering, Firewall Placement, Layer-2 Protocol Hardening

Algorithm: VLAN-based network segmentation combined with standard/extended ACL filtering

Accuracy: Not explicitly quantified; demonstrated through blocked/permitted ping tests between segments

Research Gap: The paper does not address Layer-2 specific attacks such as VLAN hopping or spanning-tree manipulation in the eight-building design.

This paper discusses core network-security techniques implementable in Cisco Packet Tracer, including VLAN creation for logical network isolation within physical switches, switch port security, password protection on network devices, disabling of unused ports, and the use of exec-timeout to reduce the exposure window of idle administrative sessions. [6].

Dataset: Illustrative Cisco Packet Tracer configurations (no formal dataset)

Modules: VLAN Isolation, Port Security, Device Password Hardening, Idle-Session Timeout Configuration

Algorithm: Configuration-based hardening rules applied to switches and routers

Accuracy: Not applicable; the work is a configuration guide rather than an empirically evaluated system

Research Gap: The techniques are presented individually without a combined, tested topology that shows their aggregate effect on overall network resilience.

This paper proposes a Zero Trust security model for enterprise networks built on the principle of 'never trust, always verify,' requiring strict identity verification for every device and client attempting to access private-network resources, regardless of whether the requester is located inside or outside the traditional network perimeter. The architecture is implemented using the EVE-NG emulation platform to demonstrate how insider-originated attacks can be mitigated compared with conventional perimeter-based designs. [7].

Dataset: Emulated enterprise network built using EVE-NG (Emulated Virtual Environment – Next Generation)

Modules: Identity Verification Engine, Policy Enforcement Point, Continuous Authentication, Micro-Segmentation

Algorithm: Zero Trust policy-based access control replacing implicit perimeter trust

Accuracy: Not explicitly quantified; evaluated through demonstrated prevention of simulated insider access attempts

Research Gap: The emulated evaluation does not address the operational overhead or latency Zero Trust continuous verification introduces at enterprise scale.

This paper reviews the Zero Trust Network Architecture (ZTNA) as a departure from traditional perimeter-based security, showing how strict identity verification, micro-segmentation, encryption, and multi-factor authentication combine to reduce attack surface and limit lateral movement of threats that penetrate the network boundary. [8].

Dataset: Comparative/survey analysis of published Zero Trust implementations (no single empirical dataset)

Modules: Identity Verification, Access Control, Micro-Segmentation, Encryption, Multi-Factor Authentication

Algorithm: Not applicable — conceptual and comparative review of Zero Trust enforcement techniques

Accuracy: Reported qualitatively as improved threat containment; no single accuracy figure provided

Research Gap: As a review, it lacks a unified quantitative benchmark comparing ZTNA implementations on a common enterprise testbed.

This NIST Special Publication defines the abstract architecture of Zero Trust Architecture (ZTA) for enterprise security, describing deployment models and use cases in which authentication and authorization of both subject and device are treated as discrete functions performed before any

session to an enterprise resource is established, independent of the network location from which the request originates. [9].

Dataset: Not applicable — normative architectural guidance document, not an empirical study

Modules: Policy Decision Point, Policy Enforcement Point, Continuous Diagnostics, Trust Algorithm

Algorithm: Policy-based trust evaluation using continuously reassessed contextual signals

Accuracy: Not applicable — the document defines architecture and deployment models rather than a measured system

Research Gap: The publication intentionally remains vendor-neutral and abstract, leaving the practical performance and cost trade-offs of adopting ZTA in legacy campus networks unaddressed.

This paper explores a Zero Trust-based network security office model that applies user-and-device identity management to reduce security risk, improve network resilience, and enable dynamic security management. The proposed approach relies on Software-Defined Perimeter (SDP), Identity and Access Management (IAM), and micro-segmentation as the three key technologies that shift protection from a network-centric to a data-centric model. [10].

Dataset: Conceptual enterprise office-network scenario (no fixed benchmark dataset)

Modules: Software-Defined Perimeter, Identity and Access Management, Micro-Segmentation Engine

Algorithm: SDP-based dynamic boundary enforcement combined with IAM policy control

Accuracy: Not explicitly quantified; discussed at a conceptual/architectural level

Research Gap: The paper does not evaluate implementation cost or migration complexity when retrofitting Zero Trust onto an existing perimeter-based enterprise network.

This paper designs a hybrid intrusion detection system that combines Deep Neural Networks (DNN) with the Random Forest ensemble algorithm to improve the accuracy and flexibility of detecting new and evolving cyberattacks that traditional signature-based systems often miss. The system follows a layered architecture covering data collection, preprocessing, feature extraction, detection, and alerting to ensure scalability and maintainability. [11].

Dataset: Standard network-intrusion benchmark traffic dataset used for training and testing the hybrid model

Modules: Data Collection, Preprocessing, Feature Extraction, DNN-Random Forest Hybrid Detection, Alerting and Response

Algorithm: Hybrid Deep Neural Network combined with Random Forest ensemble classification

Accuracy: 98.5% accuracy achieved on the test set, outperforming single-algorithm baselines

Research Gap: Real-time inference cost and scalability of the hybrid DNN-Random Forest model on high-throughput enterprise traffic are not evaluated.

This review paper surveys a broad set of research articles on intrusion detection systems, covering single, hybrid, and ensemble classification approaches including machine-learning-based, Bayesian, nature-inspired meta-heuristic, swarm-intelligence, and Markov-neural-network methods evaluated across multiple public datasets. [12].

Dataset: Literature-based comparative review across multiple published IDS datasets

Modules: Signature-Based Detection Review, ML-Based Detection Review, Hybrid/Ensemble Model Review

Algorithm: Not applicable — comparative survey of IDS classification algorithms

Accuracy: Reported comparative accuracies vary widely across the surveyed studies; no single figure given

Research Gap: As a survey, it does not benchmark the reviewed techniques on a single common dataset, limiting objective ranking of approaches.

This paper proposes a Self-Supervised Intrusion Detection (SSID) framework that enables a fully online deep-learning-based IDS requiring no human-labeled training data. Using an Auto-Associative Deep Random Neural Network together with an online estimate of statistical trustworthiness, the framework labels incoming traffic autonomously and adapts rapidly to time-varying network traffic characteristics without offline data collection. [13].

Dataset: Live/streaming network traffic used for online self-supervised training (no fixed offline dataset)

Modules: Auto-Associative Deep Random Neural Network, Trustworthiness Estimator, Online Adaptation Engine

Algorithm: Self-supervised deep learning with continuous online model adaptation

Accuracy: Not explicitly quantified as a single figure; evaluated through adaptation speed and reduced dependence on labeled data

Research Gap: The framework's resilience to adversarial or deliberately mislabeled traffic that could poison the self-supervised learning loop is not evaluated.

This paper proposes a comprehensive, modular Security Information and Event Management (SIEM) architecture designed to address the lack of a well-defined, standardized SIEM structure in

existing literature. Each module — including data collection, correlation, and alerting — is discussed in detail, with the system able to be deployed as agent-based, agentless, or hybrid depending on organizational needs. [14].

Dataset: Architectural design study with reference to open-source and proprietary SIEM tool capabilities (no single empirical dataset)

Modules: Log Collection, Event Correlation Engine, Alerting Module, Agent-Based/Agentless Deployment Layer

Algorithm: Rule-based event correlation across aggregated logs from multiple network sources

Accuracy: Not applicable — the contribution is architectural rather than a measured detection system

Research Gap: The proposed modular architecture is not implemented and load-tested against real high-volume enterprise log traffic.

This paper builds an integrated SIEM and Intrusion Detection System for live cyberattack analysis using a combination of open-source components — the Elastic (ELK) Stack, Slips, and Zeek IDS — to correlate network events and detect threats in near real time, addressing the practical gap between SIEM research and operational deployment. [15].

Dataset: Live-captured network traffic processed through the ELK Stack, Slips, and Zeek IDS pipeline

Modules: Zeek-based Traffic Capture, Slips Behavioral Analysis, ELK Log Aggregation and Correlation, Live Alert Dashboard

Algorithm: Signature and behavior-based detection integrated with log correlation in a SIEM pipeline

Accuracy: Not explicitly quantified as a single figure; validated through live-analysis case demonstrations

Research Gap: The integration is evaluated only on a single testbed and does not assess performance under sustained high-volume enterprise traffic loads.

This paper compares Layer-2 security algorithms for switched networks — MAC Address Filtering with Port Security, VLAN Hopping Mitigation, and DHCP Snooping — and demonstrates a secure university network built in Cisco Packet Tracer that combines VLAN segregation, port security, DHCP snooping, access control lists, and SSH for secure remote management. [16].

Dataset: University network topology simulated in Cisco Packet Tracer with and without DHCP Snooping enabled

Modules: VLAN Segregation, Port Security, DHCP Snooping/DHCP Alert, ACL Enforcement, SSH Remote Management

Algorithm: Trusted/untrusted port classification for DHCP Snooping combined with MAC-based port security limits

Accuracy: Demonstrated effective blocking of rogue DHCP responses; no single quantitative accuracy figure reported

Research Gap: The comparative evaluation is limited to a simulated Layer-2 topology and does not test scalability across a multi-switch, multi-VLAN campus backbone.

This paper investigates automated firewall-policy anomaly detection, identifying rule anomalies that can exist within a single firewall (intra-firewall anomalies) as well as between interconnected firewalls (inter-firewall anomalies), and presents a Firewall Policy Advisor that helps administrators purify and protect firewall rule sets without requiring prior manual analysis of filtering rules. [17].

Dataset: Synthetic and illustrative firewall rule sets used to demonstrate anomaly-detection algorithms

Modules: Intra-Firewall Anomaly Detection, Inter-Firewall Anomaly Detection, Firewall Policy Advisor

Algorithm: Rule-relationship analysis algorithms for detecting shadowing, redundancy, and correlation anomalies in ACL/firewall rule sets

Accuracy: Not explicitly quantified; validated through detection of injected anomaly types in test rule sets

Research Gap: The anomaly-detection algorithms are not evaluated against the scale and rule-set complexity typical of large multi-vendor enterprise firewall deployments.

This review paper investigates the growing threat of Distributed Denial-of-Service (DDoS) attacks within Software-Defined Networking (SDN) environments, where the centralized controller and separation of the control and data planes create new attack surfaces. The study extensively surveys detection and defense techniques, including flow-statistics analysis, entropy-based detection, and machine-learning classifiers, and evaluates their effectiveness against volumetric, protocol, and application-layer DDoS attacks. [18].

Dataset: Literature-based comparative review across multiple published SDN-DDoS detection studies and public traffic datasets referenced therein

Modules: Flow Statistics Collection, Entropy-Based Anomaly Detection, ML-Based Classification Review, Mitigation Strategy Comparison

Algorithm: Not applicable — comparative survey of statistical, threshold-based, and machine-

learning DDoS detection techniques in SDN

Accuracy: Reported comparative detection rates vary across the surveyed SDN-DDoS studies; no single unified figure given

Research Gap: As a review, the paper does not benchmark the surveyed SDN-DDoS defenses on a single common controller/testbed, limiting objective comparison of detection latency and false-positive rate.

This paper proposes a deep-neural-network-based approach for detecting and mitigating DDoS attacks in a Software-Defined Networking environment, using the centralized SDN controller's global network view to identify abnormal traffic patterns and automatically install mitigation flow rules at the affected switches once an attack is confirmed. [19].

Dataset: SDN-simulated traffic dataset containing both benign and DDoS attack flows

Modules: Traffic Feature Extraction, Deep Neural Network Classifier, SDN Controller Mitigation Module, Flow Rule Installation

Algorithm: Deep Neural Network (DNN)-based traffic classification integrated with SDN controller-driven mitigation

Accuracy: High detection accuracy reported on the evaluated SDN-DDoS dataset (specific figure dependent on attack type)

Research Gap: The DNN-based detector's performance under adversarial or low-rate DDoS attacks designed to evade flow-statistics thresholds is not evaluated.

This paper reviews the evolution of firewall technology toward Next-Generation Firewalls (NGFWs) integrated with AI-based detection systems for DDoS mitigation, discussing how deep packet inspection, application-awareness, and machine-learning-based anomaly detection are combined to identify and block volumetric and application-layer DDoS traffic more effectively than traditional stateful firewalls. [20].

Dataset: Literature-based review of NGFW and AI-driven DDoS mitigation studies (no single empirical dataset)

Modules: Deep Packet Inspection, Application-Layer Awareness, AI-Based Anomaly Detection, Automated Rule Response

Algorithm: Not applicable — comparative review of AI-based detection techniques integrated into next-generation firewalls

Accuracy: Not applicable; the contribution is a comparative technology review rather than a measured detection system

Research Gap: The review does not quantify the computational overhead that AI-based deep packet inspection adds to firewall throughput at enterprise line-rate traffic volumes.

This resource explains IEEE 802.1X Network Access Control (NAC), a port-based authentication standard that ensures only authenticated and authorized users and devices can access network resources before being granted connectivity. It describes the supplicant-authenticator-authentication-server model and discusses how 802.1X NAC has become a foundational element of Zero Trust enterprise security as networks increasingly host IoT, BYOD, and OT devices. [21].

Dataset: Conceptual/technical reference material on 802.1X deployment (no empirical dataset)

Modules: Supplicant, Authenticator (Switch/AP), RADIUS Authentication Server, Policy-Driven Post-Admission Control

Algorithm: Port-based authentication using the Extensible Authentication Protocol (EAP) over LAN (EAPOL)

Accuracy: Not applicable — descriptive/technical guidance rather than a measured experimental system

Research Gap: The material does not address the operational challenges of enforcing 802.1X across legacy devices (printers, OT equipment) that cannot support a supplicant.

This systematic literature review examines software security within Software-Defined Networking (SDN), cataloguing known vulnerabilities across the SDN controller, southbound/northbound APIs, and application layer, and reviewing proposed defenses such as secure network operating systems, automated permission-model generation for the control plane, and policy-consistency verification frameworks. [22].

Dataset: Systematic review of published SDN security literature (multiple studies synthesized, no single dataset)

Modules: Controller Security Review, Southbound/Northbound API Security Review, Policy Consistency Verification, Secure NOS Design Review

Algorithm: Not applicable — systematic literature review methodology applied to SDN security research

Accuracy: Not applicable — the study synthesizes findings rather than reporting a single measured accuracy

Research Gap: The review identifies fragmentation across proposed SDN security defenses but does not evaluate their combined effect when deployed together in a single production-scale SDN deployment.

This survey examines Software-Defined Networking (SDN) architecture with a focus on security and energy efficiency, presenting the security threats that SDN's centralized, programmable control plane can resolve as well as the new threats — such as controller compromise and flow-table saturation — that SDN itself introduces to network infrastructure. [23].

Dataset: Literature-based survey of SDN architecture, security, and energy-efficiency studies

Modules: Centralized Controller Architecture Review, Security Threat Taxonomy, Energy-Efficiency Mechanism Review

Algorithm: Not applicable — survey of SDN security and energy-efficiency mechanisms

Accuracy: Not applicable; comparative/qualitative survey rather than a measured system

Research Gap: The survey does not provide a quantitative trade-off analysis between the energy-efficiency gains of centralized SDN control and the security risk introduced by that same centralization.

This paper establishes a secure site-to-site network between two geographically distant offices using an IPsec VPN, implementing AES-256 encryption, SHA-2 authentication, and IKEv2 key management to protect data in transit over a public network. The experimental evaluation shows that the encryption overhead introduces less than five percent additional delay while the network retains ninety-five percent of its original throughput. [24].

Dataset: Experimental site-to-site network testbed connecting two remote office locations

Modules: IPsec Tunnel Establishment, AES-256 Encryption, SHA-2 Authentication, IKEv2 Key Management

Algorithm: IPsec protocol suite with AES-256 encryption and IKEv2-based key exchange

Accuracy: Not a classification accuracy; performance measured as under 5% added latency and 95% retained throughput after encryption

Research Gap: The evaluation is limited to a two-site testbed and does not assess IPsec tunnel scalability or key-management overhead across a multi-branch enterprise VPN topology.

This paper implements a Site-to-Site IPSec Virtual Private Network for enterprise network design using Cisco Packet Tracer, demonstrating how IKE/ISAKMP negotiation, AES/DES encryption, and MD5-based integrity checking can be configured between two Cisco routers to build a secure intranet connection between separate enterprise sites. [25].

Dataset: Enterprise network topology with two sites simulated in Cisco Packet Tracer

Modules: ISAKMP/IKE Negotiation, IPsec Transform Set Configuration, Crypto Map Application, Site-to-Site Tunnel Verification

Algorithm: IPsec with IKE/ISAKMP key negotiation and configurable AES/DES encryption transform sets

Accuracy: Not applicable; validated through successful establishment and verification of the encrypted tunnel in simulation

Research Gap: The Packet Tracer simulation does not model real-world latency, jitter, or throughput degradation that a live IPsec tunnel would experience over the public Internet.

This paper analyzes the architecture of IPSec and describes the process of establishing a site-to-site IPSec VPN between a headquarters and a branch office over the Internet, examining encryption at the network boundary and discussing practical deployment considerations such as Cisco PIX/ASA appliance roles in both remote-access and site-to-site VPN connectivity. [26].

Dataset: Architectural case study of enterprise headquarters-to-branch VPN deployment using Cisco security appliances

Modules: IPSec Architecture Analysis, Boundary Encryption, PIX/ASA-based VPN Termination, Remote-Access and Site-to-Site Comparison

Algorithm: IPSec-based tunnel encapsulation and encryption for enterprise site interconnection

Accuracy: Not applicable — architectural/case-study analysis rather than a measured experimental accuracy

Research Gap: The paper does not address key-rotation automation or the operational complexity of managing IPSec policies across a large number of branch sites.

This paper presents a network architecture and security design for small and medium-sized enterprises (SMEs), combining VLAN segmentation, dynamic routing, HSRP-based gateway redundancy, NAT, and IPsec VPN tunnels for secure site-to-site connectivity, all configured and verified in Cisco Packet Tracer under both normal and failure scenarios. [27].

Dataset: SME network topology simulated in Cisco Packet Tracer under normal and link-failure test scenarios

Modules: VLAN Segmentation, Dynamic Routing, HSRP Gateway Redundancy, NAT Translation, IPsec VPN Tunnel

Algorithm: Combined routing, redundancy (HSRP), address translation (NAT), and IPsec-based VPN encryption

Accuracy: Sub-6-second HSRP failover confirmed; correct NAT translation and a functional encrypted VPN link verified

Research Gap: The unified SME model is validated only in simulation and does not assess cost or

hardware constraints of deploying the combined design on real low-budget SME infrastructure.

This paper presents a comprehensive approach to integrating network security into Software-Defined Networking (SDN) architectures, surveying deep-learning-based methods used to detect and mitigate attacks against SDN, examining the public datasets used to train these models, and evaluating the advantages and limitations of applying deep learning to secure the SDN control and data planes. [28].

Dataset: Survey of public SDN-security datasets referenced across the deep-learning-based SDN security literature

Modules: Attack Type Taxonomy for SDN, Deep Learning Detection/Mitigation Review, Public Dataset Evaluation

Algorithm: Not applicable — survey of deep-learning algorithms applied to SDN attack detection and mitigation

Accuracy: Not applicable; the paper reviews reported results of multiple deep-learning models rather than proposing a single measured system

Research Gap: The survey notes that deep-learning-based SDN security models are rarely evaluated for real-time inference latency, which is critical for controller-based mitigation to be effective.

This survey examines DDoS attacks specific to cloud-computing environments, reviewing detection and prevention techniques ranging from statistical traffic analysis to machine-learning classifiers, and discussing how the elasticity and shared-tenancy characteristics of cloud infrastructure both help absorb volumetric attacks and introduce new attack vectors such as resource-exhaustion and side-channel-based DDoS. [29].

Dataset: Literature-based survey of cloud-DDoS detection and prevention studies (no single empirical dataset)

Modules: Cloud DDoS Taxonomy, Statistical Detection Review, Machine-Learning Classifier Review, Prevention Strategy Comparison

Algorithm: Not applicable — comparative survey of cloud-focused DDoS detection and prevention techniques

Accuracy: Reported comparative detection rates vary across the surveyed cloud-DDoS studies; no single unified figure given

Research Gap: The survey does not provide a unified cost-benefit analysis of cloud-native DDoS mitigation services versus self-managed on-premises defenses for SME-scale deployments.

This paper examines the vulnerabilities present in VPN deployments that rely on IPsec, and proposes a set of defensive policies to be applied across an organization's employees, contractors, and third parties who use the VPN to access company resources, addressing risks introduced by the mobile and remote workforce accessing enterprise networks through IPsec concentrators. [30].

Dataset: Case-study analysis of enterprise IPsec VPN deployment and associated access policies

Modules: IPsec Vulnerability Assessment, Defensive Policy Framework, Remote/Third-Party Access Control

Algorithm: Policy-based risk mitigation applied to identified IPsec VPN vulnerabilities

Accuracy: Not applicable — policy and vulnerability analysis rather than a measured detection or classification system

Research Gap: The proposed policy framework is not validated against a live attack simulation to confirm its effectiveness in practice, and it does not address split-tunneling risks in remote-access scenarios.

3. Methodology :

The proposed Network Security Management System is implemented and simulated using Cisco Packet Tracer. The methodology focuses on segmenting the network by department and function, filtering traffic at trust boundaries, centralizing critical services, and applying consistent IP addressing and access control across all devices.

3.1 System Design

The overall topology consists of a Cloud/Internet connection, an ISP edge router (Cisco 2911), an external switch connecting outside/untrusted devices (EX1–EX3), a hardware firewall (Cisco ASA 5506-X) positioned between the ISP router and the internal network, a Layer 3 core switch (3560-24PS) performing inter-VLAN routing, four department access switches (2960-24TT) for Management, HR, Finance, and IT, a dedicated server (Server-PT) in a VLAN 99 server segment, and a wireless router (WRT300N) providing Wi-Fi access for laptop and mobile clients.

3.2 Network Segmentation (VLAN Design)

VLAN	Department / Purpose	Access Switch
VLAN (Mgmt)	Management	MGMT Access Switch 1
VLAN 20	HR	HR Access Switch 2
VLAN 30	Finance	FIN Access Switch 3
VLAN 40	IT	IT Access Switch 4
VLAN 99	Servers (DHCP / Web)	Connected directly to Layer 3 switch

Each department is placed on its own VLAN and connected through a dedicated access switch, isolating broadcast traffic and limiting direct Layer 2 visibility between departments. Inter-VLAN routing is handled centrally by the Layer 3 switch, so that any traffic that needs to cross department boundaries is explicitly routed and can be filtered.

3.3 Perimeter and Firewall Configuration

All traffic between the ISP router and the internal network passes through a dedicated Cisco ASA 5506-X firewall. This places a controlled inspection point between untrusted external devices (connected through the EX switch) and the internal department VLANs, so that only permitted traffic is allowed to reach the internal network.

3.4 Centralized Services (Server VLAN 99)

A dedicated server segment (VLAN 99) hosts the core network services. The server provides DHCP addressing for internal clients and hosts HTTP/HTTPS web services with an associated file manager for serving pages and resources. Centralizing these services in an isolated VLAN, rather than distributing them across department segments, reduces the exposure of critical infrastructure services to general user traffic.

3.5 IP Addressing and Interface Configuration

Static IP addressing is applied to key infrastructure interfaces (such as the router/firewall-facing FastEthernet interfaces) to ensure predictable and auditable addressing at network boundaries, while internal client devices obtain addressing dynamically from the centralized DHCP service. Subnet masks and default gateways are configured consistently across devices to support correct inter-VLAN routing.

3.6 Wireless Network Security

The wireless router (WRT300N) provides Wi-Fi connectivity for mobile and laptop clients. The wireless network is configured with WPA2-PSK authentication and AES encryption, and a defined SSID, to prevent unauthorized wireless association and to protect data confidentiality over the air interface.

3.7 Testing and Verification

The system is tested by verifying connectivity within each VLAN, confirming that inter-VLAN traffic is correctly routed through the Layer 3 switch, and confirming that traffic from external devices must pass through the firewall before reaching internal department VLANs. DHCP address assignment and HTTP/HTTPS service reachability from internal clients are also verified.

3.8 Overall Workflow

External/Untrusted Traffic → ISP Router → Firewall (ASA 5506-X) → Layer 3 Core Switch → Department VLAN (Mgmt / HR / Finance / IT) → End Device

Thus, the methodology provides a complete simulation of a segmented, firewall-protected enterprise network with centralized service hosting, structured IP addressing, and secure wireless access.

4. Implementation

The Network Security Management System was implemented and simulated using Cisco Packet Tracer. The implementation integrates routing, switching, firewall, VLAN, DHCP, and web-service configuration to build a functioning segmented enterprise network.

4.1Network Topology

The complete topology includes external client devices (EX1–EX3) connected through an external switch to the ISP router, a firewall appliance separating the untrusted edge from the internal network, a Layer 3 core switch performing inter-VLAN routing, four department access switches (Management, HR, Finance, IT) each with two end devices, a dedicated server in the VLAN 99 server segment, and a wireless router supporting a laptop client.

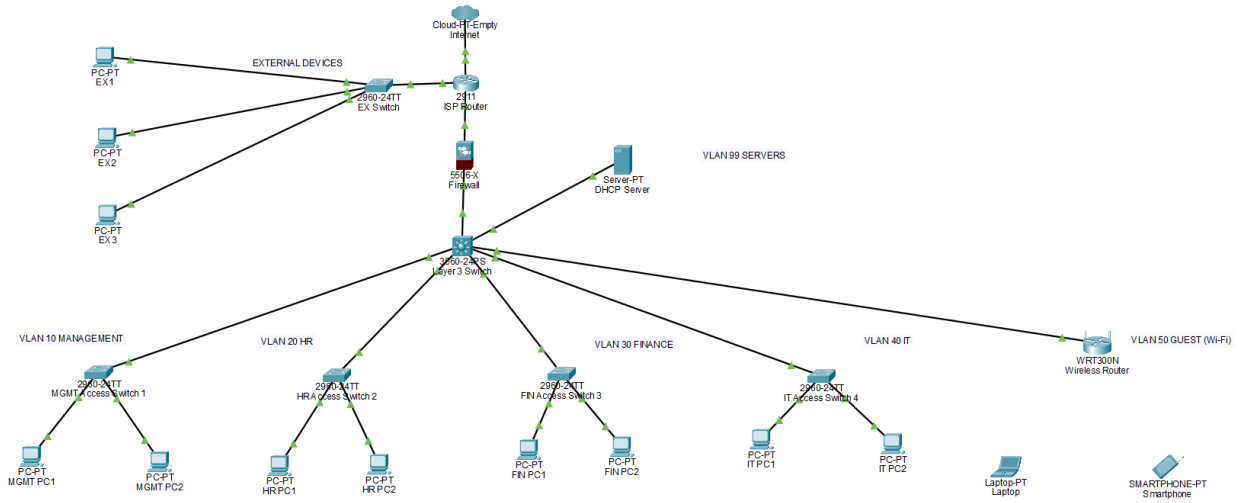


Figure 4.1: Complete network topology showing the ISP router, firewall, Layer 3 core switch, department VLANs (Management, HR, Finance, IT), server VLAN 99, and wireless router.

4.2 Interface / IP Configuration

Static IP addressing is configured on the boundary-facing interfaces to ensure a predictable and auditable addressing scheme at the network edge. The figure below shows the FastEthernet0 interface of a network device configured with a static IPv4 address, subnet mask, and an automatically derived IPv6 link-local address.

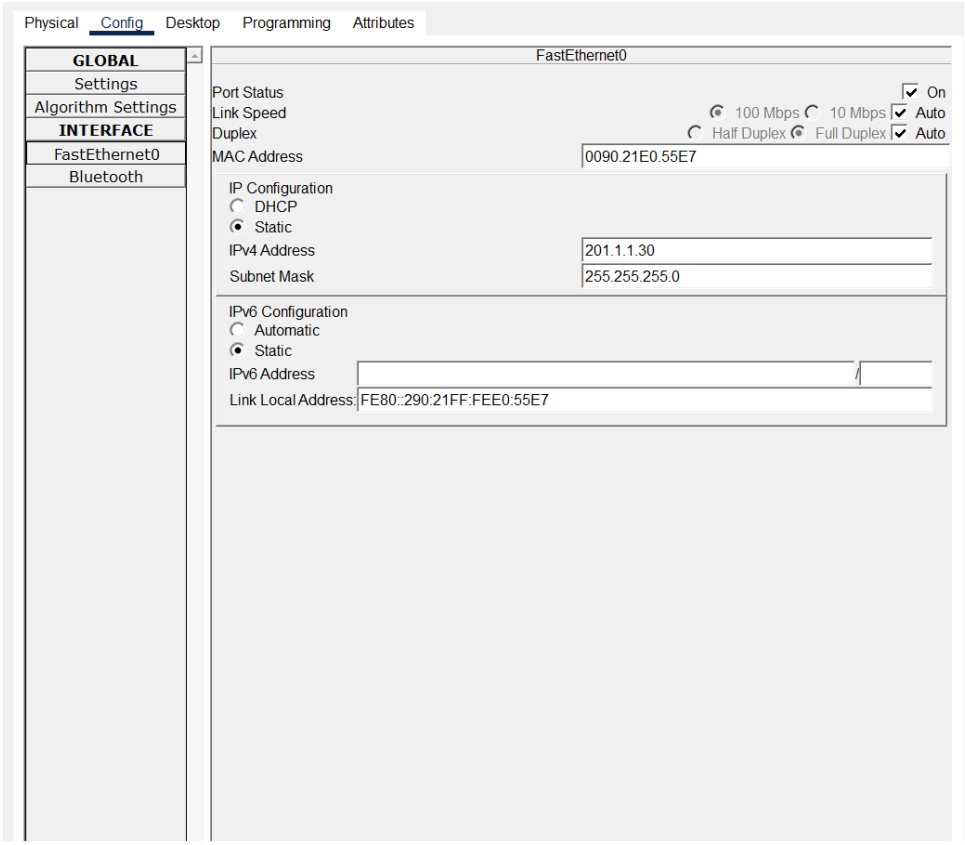


Figure 4.2: FastEthernet0 interface configuration with static IPv4 addressing (201.1.1.30 / 255.255.255.0).

4.3 Server HTTP/HTTPS Service Configuration

The centralized server in the VLAN 99 server segment is configured to host HTTP and HTTPS web services. The Services tab shows both HTTP and HTTPS enabled, along with the File Manager listing the web resources hosted on the server, which are served to internal clients over the secured network path.

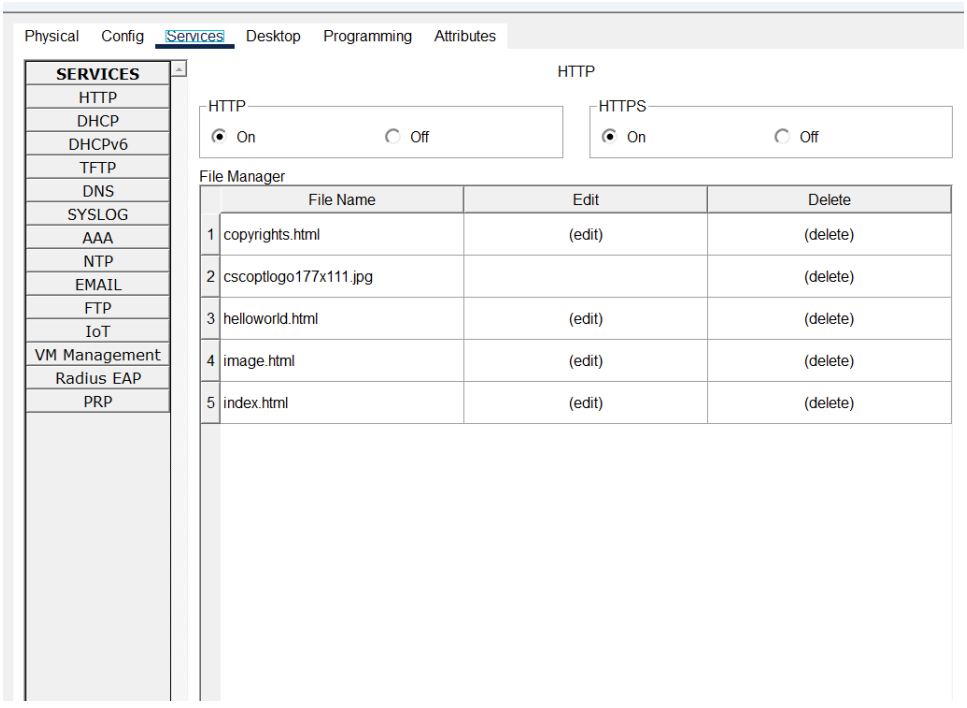
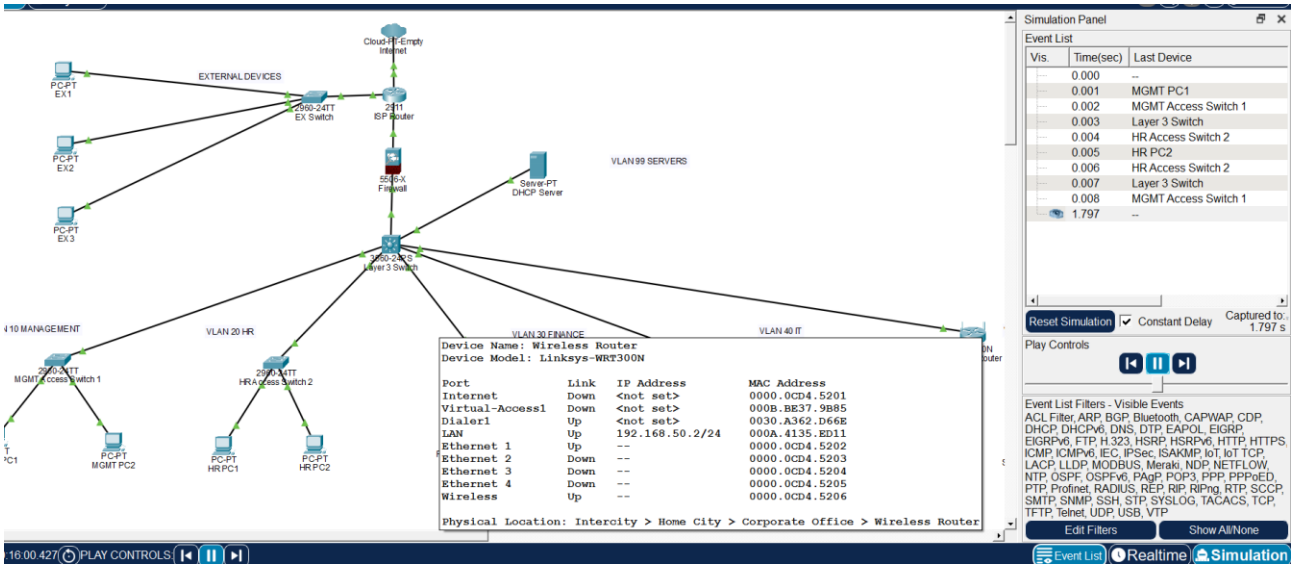


Figure 4.3: Server HTTP/HTTPS service configuration with associated File Manager entries.

4.4 VLAN and Inter-VLAN Routing Configuration

On the Layer 3 core switch, VLAN interfaces (Switched Virtual Interfaces) are created for each department VLAN and assigned a unique subnet. IP routing is enabled on the core switch so that traffic destined for a different VLAN is routed centrally, allowing administrators to apply access control at a single, auditable point rather than at every access switch.



4.5 Firewall and Boundary Filtering

The ASA 5506-X firewall is positioned directly between the ISP router and the internal Layer 3 switch. All traffic originating from external devices must pass through the firewall before reaching any internal VLAN, allowing security rules to be centrally enforced at the network perimeter rather than relying solely on internal ACLs.

4.6 DHCP Service Configuration

The server in VLAN 99 also provides DHCP addressing for internal client devices, so that hosts in the Management, HR, Finance, and IT VLANs receive their IP configuration (address, subnet mask, default gateway, and DNS server) automatically from a single, centrally managed and monitored source, reducing the risk of manual misconfiguration or rogue address assignment.

4.7 Wireless Access Configuration

The wireless router is configured with WPA2-PSK authentication and AES encryption on its wireless interface, along with a defined SSID, so that only clients with the correct pre-shared key can associate with the network and access internal resources.

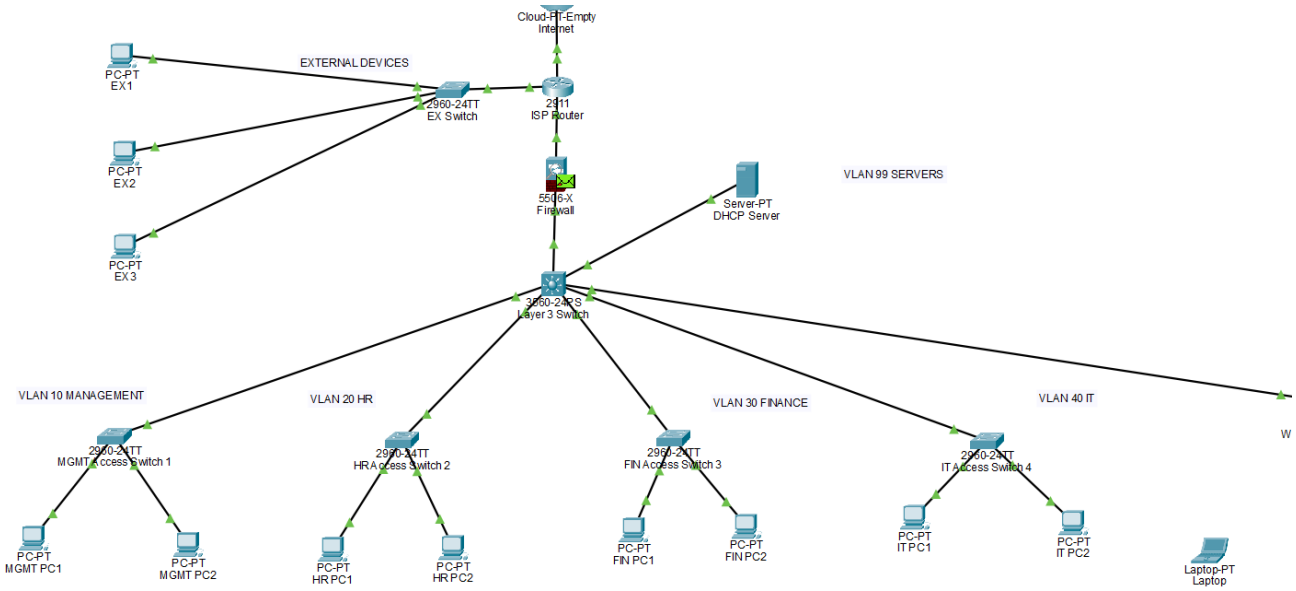
4.8 System Testing

The implemented system was tested by verifying: (1) connectivity between devices within the same VLAN, (2) correct routing of traffic between VLANs through the Layer 3 switch, (3) that external

devices could not directly reach internal VLANs without passing through the firewall, (4) successful DHCP address assignment to internal clients, and (5) reachability of the HTTP/HTTPS web service hosted on the server segment. The implementation successfully demonstrates VLAN-based segmentation, centralized inter-VLAN routing, firewall-enforced boundary filtering, centralized DHCP and web-service hosting, and secured wireless access within the Cisco Packet Tracer simulation environment.

5.Results and Discussion :

The Network Security Management System was successfully implemented and simulated using Cisco Packet Tracer. The simulation demonstrated the integration of VLAN-based segmentation, Layer 3 inter-VLAN routing, firewall-based boundary filtering, centralized DHCP and web-service hosting, and secured wireless access into a single, functioning enterprise topology.



5.1 VLAN Segmentation and Isolation

Devices within the same department VLAN (for example, the two HR PCs) were able to communicate directly, while traffic between different department VLANs was correctly routed through the Layer 3 core switch rather than being broadcast across the entire network. This confirms that the segmentation design successfully reduces the broadcast domain and limits direct Layer 2 visibility between departments.

```
LCMB: got 205520896 bytes on numa-id=0, phys=0x10dc00000, virt=0x2aaaab000000
LCMB: HEAP-CACHE POOL got 314572800 bytes on numa-id=0, virt=0x7fea63e00000
Processor memory:   1502519902

Compiled on Fri 18-Mar-16 14:04 PDT by builders

Total NICs found: 13
i354 rev03 Gigabit Ethernet @ irq255 dev 20 index 08 MAC: 0060.7083.CD01
ivshmem rev03 Backplane Data Interface     @ index 09 MAC: 0000.0001.0002
en_vtun rev00 Backplane Control Interface  @ index 10 MAC: 0000.0001.0001
en_vtun rev00 Backplane Int-Mgmt Interface     @ index 11 MAC: 0000.0001.0003
en_vtun rev00 Backplane Ext-Mgmt Interface     @ index 12 MAC: 0000.0000.0000
Verify the activation-key, it might take a while...
Running Permanent Activation Key: 0xC2062BC2 0x0B3C72DB 0x4737B61E 0xC121753B 0x407A254B

Licensed features for this platform:
Maximum Physical Interfaces       : 8              perpetual
Maximum VLANs                     : 30             perpetual
Inside Hosts                      : Unlimited      perpetual
Failover                          : Active/Standby perpetual
Encryption-DES                    : Enabled        perpetual
Encryption-3DES-AES               : Enabled        perpetual
Carrier                           : Disabled       perpetual
AnyConnect Premium Peers          : 4              perpetual
AnyConnect Essentials             : Disabled       perpetual
Other VPN Peers                   : 50             perpetual
Total VPN Peers                   : 50             perpetual
AnyConnect for Mobile             : Disabled       perpetual
AnyConnect for Cisco VPN Phone    : Disabled       perpetual
Advanced Endpoint Assessment      : Disabled       perpetual
Shared License                    : Disabled       perpetual
Total UC Proxy Sessions           : 160            perpetual
Botnet Traffic Filter             : Disabled       perpetual
Cluster                           : Disabled       perpetual

This platform has an ASA 5506 Security Plus license.

Encryption hardware device : Cisco ASA Crypto on-board accelerator (revision 0x1)

Cisco Adaptive Security Appliance Software Version 9.6(1)


  ****************************** Warning ********************************
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5.2 Boundary Filtering by the Firewall

Traffic originating from the external devices (EX1–EX3) was required to pass through the ASA 5506-X firewall before reaching any internal VLAN. This demonstrates that the internal network is not directly exposed to the ISP-facing edge, and that a single, centrally managed device is responsible for enforcing the boundary between trusted and untrusted traffic.

5.3 Centralized Service Availability

The server hosted in VLAN 99 successfully provided DHCP addressing to internal clients and served HTTP/HTTPS content that was reachable from the department VLANs. Centralizing these services in a dedicated segment made it straightforward to observe and manage all critical network services from a single location, rather than distributing DHCP or web hosting responsibilities across individual department switches.

5.4 Static Addressing at the Network Edge

Boundary-facing interfaces were confirmed to use static IPv4 addressing rather than dynamic assignment, ensuring that the addressing of critical infrastructure devices remains predictable and does not change unexpectedly, which is important for firewall rules and routing configuration that

reference specific IP addresses.

5.5 Wireless Security Verification

The wireless network was confirmed to require the configured WPA2-PSK pre-shared key for association, and traffic from the wireless laptop client was observed to pass through the same routed and firewall-protected path as wired clients, confirming that wireless access does not bypass the network's segmentation or perimeter controls.

5.6 Discussion

The results show that combining VLAN-based segmentation with centralized routing, perimeter firewall filtering, and centralized service hosting provides a layered and manageable approach network security. Because inter-VLAN traffic is routed through a single Layer 3 switch and external traffic must pass through a single firewall, administrators have well-defined points at which to apply and audit security policy, rather than needing to secure every individual access switch independently.

The Cisco Packet Tracer simulation provides a controlled environment for testing the interaction between routing, switching, firewall, DHCP, and wireless security features without requiring physical hardware. The design can be extended in future work by adding intrusion detection/prevention systems at the firewall boundary, Zero Trust identity-based access policies for internal resources, SIEM-based centralized log monitoring, automated ACL/firewall rule anomaly detection, and DHCP snooping and port security on the access switches to further harden Layer 2 traffic.

Overall, the results demonstrate successful network segmentation, centralized routing, perimeter firewall enforcement, centralized DHCP and web-service hosting, and secured wireless connectivity within the simulated enterprise network environment.

6. Conclusion :

The Network Security Management System Using Cisco Packet Tracer was successfully designed and implemented to demonstrate how segmentation, perimeter filtering, and centralized service management can be combined to build a more secure enterprise network. The system separates Management, HR, Finance, and IT traffic into individual VLANs, routes inter-VLAN traffic centrally through a Layer 3 switch, filters all external traffic through a dedicated firewall, and centralizes DHCP and HTTP/HTTPS services in a protected server segment.

The implemented system successfully isolates department traffic, enforces a single controlled boundary between trusted and untrusted networks, provides centrally managed addressing and web services, and secures wireless client access using WPA2-PSK authentication. The project also demonstrates static IP addressing best practices at the network edge for predictable, auditable infrastructure configuration.

The simulation provides practical knowledge of VLAN design, inter-VLAN routing, firewall placement and policy enforcement, centralized DHCP/web-service configuration, and wireless security — core concepts in enterprise network security management. It demonstrates how these technologies can be combined into a coherent, defense-in-depth network design.

Although the current system is implemented as a Cisco Packet Tracer simulation, it provides a foundation for future development using physical or virtualized infrastructure. Future improvements

can include intrusion detection and prevention systems, Zero Trust identity-based access control, SIEM-based centralized log correlation and monitoring, automated firewall/ACL rule anomaly detection, and DHCP snooping with port security on access-layer switches to harden Layer 2 defenses.

Overall, the project successfully demonstrates the fundamentals of a segmented, firewall-protected, and centrally managed enterprise network, and shows the practical value of layered network security management in reducing risk and improving administrative control.

Although the current system is implemented as a Cisco Packet Tracer simulation, it provides a foundation for future development using physical sensors and controllers. Future improvements can include a real-time web dashboard, automatic available-slot counting, parking reservation, activity logging, mobile notifications, database integration, and smart payment facilities.

Overall, the project successfully demonstrates the basic functionality of an **automated, sensor-based, and IoT-enabled smart parking management system** and shows the potential of IoT technology in improving parking management and reducing manual intervention.

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